Original Article

What Predicts Attention-Deficit/Hyperactivity Disorder Literacy in Primary Schoolteachers?

Jemy J. Jome, Vidya Prabhu, Varalakshmi Chandra Sekaran, R. Vani Lakshmi[1]

Departments of Global Public Health Policy and Governance and [1]Applied Statistics and Data Sciences, Prasanna School of Public Health, Manipal Academy of Higher Education, Manipal, Karnataka, India

Abstract

Background: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common neurodevelopmental condition affecting children's learning and behaviour. Teachers play a pivotal role in early identification, yet little is known about factors predicting their ADHD literacy in India. **Aims:** This study was conducted to evaluate the Mental Health Literacy (MHL) of primary schoolteachers regarding ADHD and to identify the predictors of MHL among teachers. **Methods:** A cross-sectional, quantitative study was conducted among 303 primary schoolteachers. The study period was from December 2023 to June 2024. The study was conducted in Varkala and Attingal educational subdistricts of Thiruvananthapuram district. There are a total of 110 schools in both Attingal and Varkala subdistrict. A stratified cluster sampling technique was used to recruit the participants. Data were collected using a semi-structured validated questionnaire. Furthermore, the Knowledge of Attention Deficit Disorder Scale (KADDSs) was used. Descriptive and inferential statistics were used to assess knowledge and compare across socio-demographic characteristics. Binomial logistic regression was used to identify the predictors of MHL. **Results:** The majority of participants (56.8%) were in the age group of 35–50 years, with 94.4% being female. Most teachers (38%) had more than 10 years of teaching experience. The study found that 98.7% of teachers had a low level of knowledge related to ADHD. The mean knowledge scores showed significant association with socio-demographic characteristics such as marital status ($P = 0.01$), status of work ($P = 0.004$), type of school ($P < 0.001$), teaching hours per week ($P = 0.005$), medium of instruction ($P = 0.03$) and average class strength ($P < 0.001$). The school area and average class strength were significant predictors influencing the MHL of teachers (adjusted odds ratio [AOR]: 3.40, 95% confidence interval [CI]: 1.22–9.44; AOR = 2.38, 95% CI: 1.37–4.15, respectively). **Conclusion:** Findings show that most teachers have low ADHD knowledge, influenced by socio-demographic factors, with school area and class strength significantly predicting MHL. This highlights the need for targeted training to improve teachers' ability to support students with ADHD.

Keywords: Attention deficit/hyperactivity disorder, mental health literacy, primary school, teachers

Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is the most commonly identified childhood neurodevelopmental disease.[1] The characteristics of ADHD are problems such as lack of attention, hyperactivity and impulsiveness.[2] Inattention, impulsivity and hyperactivity are three main symptoms seen in a child who has ADHD. Lack of concentration, fidgetiness, easily distracted, forgetting or losing things a lot and difficulty in coping with others are the signs and symptoms usually exhibited by a child with ADHD.[3] If a child with ADHD is not taken care of properly, it can have several consequences throughout their life, such as academic underachievement, criminality and problems with personal relationships.[4]

World Health Organization conducted a survey which included 20 lower-to upper-income countries and revealed that the United States has the highest ADHD rate amongst children and adolescents. According to a national survey of parents using data from 2016 to 2019, an estimated 6.8% of children in the United States are diagnosed with ADHD. According to

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Address for correspondence: Dr. Vidya Prabhu, Department of Global Public Health Policy and Governance, Prasanna School of Public Health, Manipal Academy of Higher Education, Manipal, Karnataka, India.
E-mail: vidya.p@manipal.edu

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data from both the National Survey of Children's Health and the National Health Information Survey, boys have a higher chance of getting diagnosed with ADHD than girls.[5]

An umbrella review of meta-analyses reported that the global prevalence of ADHD amongst children and adolescents is approximately 8.0%.[6] Primary schoolteachers play a pivotal role in the early identification of ADHD because symptoms often emerge during the early developmental years when children begin formal education**.** Teachers are the first individuals to identify the problem and send them for treatment. At this stage, teachers observe foundational behaviours such as attention span, impulse control and social interaction core areas affected by ADHD. Given that the students spend half of their waking time in school, interacting with their teachers, teachers may act as Gateway Providers for mental health services. Teachers, as relatively neutral observers, may be better positioned than parents to identify subtle behavioural changes in children at an early stage. This is because parental observations can be influenced by emotional involvement, preconceived expectations and familiarity with the child's typical behaviour, potentially obscuring emerging symptoms.[7,8] Enhancing teachers' knowledge about ADHD promotes better mental and educational outcomes, which is linked to the objective of the United Nations Sustainable Development Goals (SDGs), particularly SDG-3 (Good Health and Well-being) and SDG-4 (Quality Education).[9] Mental health literacy (MHL) advances SDG-3 by increasing awareness, reducing stigma and promoting early intervention.[10]

Few research studies have highlighted gaps in teachers' knowledge about ADHD, which may contribute to misinterpretation of symptoms, stigma and missed opportunities for timely support and referral.[11,12]

In India, ADHD remains underdiagnosed and often misunderstood in school settings due to limited awareness, lack of formal teacher training and cultural stigma surrounding behavioural disorders. A meta-analysis of 19 Indian studies reported an ADHD prevalence ranging from 6.3% to 11.3% amongst children.[13] Although some studies have assessed teachers' knowledge of ADHD, research specifically evaluating ADHD literacy among teachers is scarce.[14-16] To address these gaps, this study was conducted to assess the ADHD literacy of primary schoolteachers and to identify predictors of MHL related to ADHD.

Conceptual framework

MHL related to ADHD in this study was assessed using a multi-method approach, incorporating knowledge of causes and treatment (via Knowledge of Attention Deficit Disorder Scale [KADDS]), recognition (via case vignettes) and behavioural indicators (training attendance, identification, referral and information-seeking behaviour). Jorm's MHL framework served as the theoretical basis for the study design[17] includes components such as problem recognition, knowledge of causes and treatments, help-seeking attitudes and information-seeking behaviour.

Independent variables were grouped into two broad categories:

- Socio-demographic variables (e.g. age, gender, marital status, education level and teaching experience) – hypothesised to influence baseline knowledge and attitudes towards ADHD
- Teaching- and work-related variables (e.g. school type, status of work, school area, teaching hours, medium of instruction and syllabus) – expected to affect exposure to training, access to resources and opportunities for identification and referral.

These variables were analysed to determine their association with ADHD-related MHL, providing a comprehensive understanding of factors influencing teachers' capacity to support students with ADHD.

Methods

Study setting and design

The study was conducted in Thiruvananthapuram district, Kerala, India. The schools in Thiruvananthapuram district are divided into 12 different educational subdistricts. The study was conducted in Varkala and Attingal educational subdistricts of Thiruvananthapuram district. There are a total of 110 schools in both Attingal and Varkala subdistricts. A cross-sectional study design was employed, and data were collected between December 2023 and June 2024.

Study population and sampling technique

The study participants were primary schoolteachers who are teaching from grades 1–7 and those teachers who are currently employed in Thiruvananthapuram district. Special education teachers are excluded from the study. The cluster stratified sampling technique was adopted. Schools were stratified into government, aided and private schools. Schools were randomly selected from each stratum, and sampling from each of these strata was performed using the probability proportionate to size [Figure 1].

Sample size

Considering 20%[18] amongst teachers with relative precision 6%, 95% confidence interval and design effect of 1.5, accounting for 20% attrition, 303 teachers were enrolled in the study.

Ethical considerations

Ethical approval was obtained from the Institutional Ethics Committee (IEC 672 - 2023). Permission was obtained from the Director General of Education (DGE), Deputy Director of Education, Kerala and respective Assistant Education Officers. Written informed consent was obtained from the participants.

Study tool

A semi-structured validated questionnaire was used to collect the data. The questionnaire comprised four sections. Sociodemographic variables, identification of adolescent mental health problems through case vignette (a brief, hypothetical scenario that describes an child with ADHD

symptoms), KADDS and general questions regarding mental health practices amongst teachers.

The KADDSs is a validated, structured questionnaire comprising 39 items designed to assess teachers' knowledge of ADHD across three key domains: symptoms and diagnosis, treatment and general information (such as its causes and courses).[19] The permission was obtained from the author to use the questionnaire. Each correct response was scored as 1, whereas incorrect or 'don't know' responses were scored as 0, with a total score range of 0–39. The KADDS score was categorised into high, moderate and low scores. Based on Bloom's cut-off, those with >80%, 60%–79% and <60% are considered to have high, moderate and low levels of knowledge, respectively.[20]

Data collection methods

Data were collected using a self-administered questionnaire. Administrative permission was obtained from the headmasters/headmistress of the respective schools before starting the data collection. After getting written consent from the participants, the questionnaire was distributed to participants and collected back after checking the completeness of the questionnaire.

Data analysis

Data were entered in Microsoft Excel, cleaned, coded and imported into Jamovi version 2.3.28[21] (The jamovi project, Sydney, New South Wales, Australia) for analysis. Descriptive analysis was done to find the frequencies and percentages. One-way analysis of variance, Mann–Whitney-*U*-test and Independent Samples – *t*-test are done to find the statistical association of mean KADDS score across the socio-demographic variables. Binomial logistic regression was done to find the predictors of MHL amongst teachers regarding ADHD.

Results

Socio-demographic characteristics of the participants.

Table 1 shows the socio-demographic characteristics of participants. Most of the participants were in the age group of 35–50 years (56.8%). Most participants were female (94.4%) and aged 34–50 years (56.8%). Most participants (47.5%) had a diploma degree in education. About 38% of participants had more than 10 years of teaching experience. The majority of the participants had more than 10 years of experience. Most educators (88.8%) were permanent employees, with 72.3% working in government institutions. 51.2% of participants taught in classes with an average strength of <30, whereas 56.8% taught for more than 10 h.

Case vignettes

About 108 (35.6%) of participants could recognise the condition as ADHD based on the case vignette. About 247 (81.5%) participants perceived that the student needed help. Out of 247 participants, about 167 (55.7%) participants opined that the student needs help from a mental health professional.

Table 1: Socio-demographic and work-related characteristics of participants (*n*=303)

Characteristics	Levels	*n* (%)
Age (years)	Up to 35	93 (30.7)
	35 to 50	172 (56.8)
	More than 50	38 (12.5)
Gender	Male	17 (5.6)
	Female	286 (94.4)
Marital status	Single	9 (3)
	Married	276 (91.1)
	Separated	18 (5.9)
Family type	Nuclear	233 (76.9)
	Joint	70 (23.1)
Education level	Diploma	144 (47.5)
	Bachelors	80 (26.4)
	Masters	79 (26.1)
Total teaching experience (years)	<1	43 (14.2)
	1–5	97 (32)
	6–10	48 (15.8)
	More than 10	115 (38)
Status of work	Permanent	269 (88.8)
	Temporary	34 (11.2)
School type	Government	219 (72.3)
	Aided	67 (22.1)
	Private	17 (5.6)
School area	Rural	279 (92.1)
	Urban	24 (7.9)
Medium of instruction	Malayalam	247 (81.5)
	English	56 (18.5)
The average strength of the class	<30	155 (51.2)
	30–40	124 (40.9)
	More than 40	24 (7.9)
Teaching hours per week	1–5	54 (17.8)
	6–10	77 (25.4)
	More than 10	172 (56.8)
Syllabus	State	290 (95.7)
	CBSE	13 (4.3)

CBSE: Central Board of Secondary Education

Mental health literacy on attention deficit hyperactivity disorder

The maximum score for the KADD scale is 39. The majority (98.7%) had a low level of knowledge, and only 1.3% had a moderate level of knowledge regarding ADHD.

The univariate analysis was carried out to assess the significant differences in the variables across groups [Table 2]. At the 5% significance level, no statistically significant difference was observed in the mean ADHD knowledge scores across age groups ($P = 0.68$), gender ($P = 0.45$), family type ($P = 0.09$), education level ($P = 0.45$), teaching experience ($P = 0.16$), school area ($P = 0.21$) and syllabus ($P = 0.06$). Statistically significant differences were found in relation to marital status ($P = 0.01$). Similarly, status of work ($P = 0.004$), type of school ($P < 0.001$), teaching hours per week ($P = 0.005$), medium of instruction ($P = 0.03$) and average strength of class ($P < 0.001$).

Table 2: Comparison of knowledge on attention-deficit/hyperactivity disorder across socio-demographic characteristics of study participants (*n*=303)

		KADDS		
Characteristics	**Levels**	**Mean±SD**	**Median (Q1–Q3)**	***P***
Age group (years)[O]	Up to 35	11.4±5.23	11 (8–15)	0.68
	35–50	11.9±4.88	12 (9–15)	
	More than 50	12.2±5.74	12 (9–16)	
Gender[I]	Female	11.7±5.04	12 (8.25–15)	0.45
	Male	12.7±6.06	13 (11–15)	
Family type[M]	Nuclear	12±5.29	12 (9–16)	0.09
	Joint	11±4.33	11 (8–14)	
Marital status[O]	Single	12.89±4.83	12 (9–15)	0.01*
	Married	11.99±5.05	12 (9–15.3)	
	Separated	8.39±4.91	9.5 (4–12.8)	
Education level[O]	Diploma	11.8±4.92	12 (9–16)	0.45
	Bachelors	12.3±5.61	12 (9–16)	
	Masters	11.3±4.88	12 (8–14)	
Teaching experience (years)[O]	<1	11.8±4.22	12 (10–14)	0.16
	1–5	11.6±4.81	11 (9–15)	
	5–10	10.5±5.50	10 (7.75–14.3)	
	More than 10	12.5±5.39	13 (9–16)	
Status of work[I]	Permanent	12.10±5.01	12 (9–16)	0.004*
	Temporary	9.41±5.20	10 (5.5–13)	
School type[O]	Government	11.06±4.81	11 (8–14)	<0.001*
	Aided	14.91±4.68	15 (12–19)	
	Private	9.06±5.34	9 (6–12)	
School area[I]	Rural	11.7±5.20	12 (8–15)	0.21
	Urban	13±3.48	13 (11–16)	
Teaching hours per week[O]	1–5	11.2±4.56	11.5 (8–13)	0.005*
	5–10	13.4±6.11	14 (9–18)	
	More than 10	11.3±4.61	11 (9–14)	
Medium of instruction[I]	Malayalam	11.5±4.94	12 (8–15)	0.03*
	English	13.1±5.61	12 (9.7–19)	
The average strength of class[O]	<30	11.17±4.90	11 (8–15)	<0.001*
	30–40	13.03±5.15	13 (10–17)	
	More than 40	9.50±4.64	10 (7–12)	
Syllabus[I]	State	11.91±5.02	12 (9–15)	0.06
	CBSE	9.23±6.13	10 (5–13)	

*P<0.05 statistically significant, [O]One-Way ANOVA, [I]Independent Samples t-test, [M]Mann–Whitney U-test. KADDS: Knowledge of Attention Deficit Disorder Scale, SD: Standard deviation, CBSE: Central Board of Secondary Education

Mental health promotion practices among participants

Table 3 depicts that about 46 (15.2%) participants had a chance to attend training related to childhood mental health problems. Among 168 (55.4) participants who identified students with behavioural issues in their career, 115 (38%) participants had referred the students to mental health professionals. About 241 (79.5%) participants had accessed various sources of information, such as TV/internet 190 (62.7%), books 34 (11.3%), health professionals 47 (15.5%), family members 17 (5.6%) and Block Resource Centre trainers 15 (4.9%).

Table 4 depicts the multivariate analysis of socio-demographic predictors on literacy related to ADHD. All the variables

Table 3: Mental health promotion practices among participants (*n*=303)

Characteristics	**Yes, *n* (%)**	**No, *n* (%)**
Attended formal training	46 (15.2)	257 (84.8)
Identified students with behavioural issues	168 (55.4)	135 (44.6)
Referred the students to a mental health professional	115 (38)	188 (62)
Information seeking behaviour	241 (79.5)	62 (20.5)

with $P < 0.25$ from the univariate binary logistic regression are taken into the multivariate regression analysis.[21] They included age group, status of work, school area, teaching hours per week and the average strength of the class. It was found that the school area and average strength of the class were the predictors of ADHD knowledge. The participants from the urban school predicted higher odds of identification of ADHD as compared to participants from rural school (adjusted odds ratio [AOR] = 3.4 [1.22, 9.44], $P = 0.01$). The participants who taught classes with a strength of 30–40 students had higher odds of identification of ADHD as compared to the participants who taught class strength of <30 students (AOR = 2.38 [1.37, 4.15], $P = 0.002$).[22]

Discussion

The current study showed that primary schoolteachers had a low level of knowledge regarding the identification of ADHD based on the vignette. A study conducted in Egypt by Elwardany *et al*. in 2015 also found that 76.1% of teachers had inadequate knowledge regarding ADHD.[23] A similar study conducted in Kerala by Fathima and Maria M reported that 62% of teachers were able to identify the presence of mental health problems in the vignette, and only 23% of teachers were able to identify the condition appropriately.[24] A cross-sectional study conducted in South India by Kashyap *et al*. reported significant gaps in schoolteachers' ability to identify and respond to student mental health issues.[25]

The current study found that most of the teachers had an overall low level of knowledge about ADHD across the scale, which comprised three subscales, including associated features, symptoms and diagnosis and treatment of ADHD. In a survey conducted by Bhattarai and Sharma in 2019 to assess knowledge of ADHD amongst schoolteachers in Nepal, similar results were obtained. The study concluded that three-fourths of the teachers had poor knowledge related to ADHD.[26]

On assessing knowledge regarding symptoms and diagnosis of ADHD, the present study found that about 143 (42.1%) teachers had moderate knowledge, but most teachers had a lower level of knowledge on both the associated features and the treatment subscale. The study conducted by Perold *et al*. in South Africa to assess the knowledge and misperceptions about ADHD also reported similar results.[27] Similarly, Saad *et al*. conducted a study in the UAE in 2022 to assess the knowledge of ADHD amongst teachers, and it was found that teachers had

Table 4: Multivariate analysis of predictors of attention-deficit/hyperactivity disorder literacy (*n*=303)

Characteristics	COR (95% CI)	*P*	AOR (95% CI)	*P*
Age group (years)				
20–35	1.14 (0.49–2.67)	0.75	1.70 (0.67–4.35)	0.26
36–50	1.96 (0.89–4.30)	0.09	2.24 (0.95–5.25)	0.06
51–65	1	-	1	
Gender				
Female	1	-		
Male	0.78 (0.28–2.11)	0.62		
Family type				
Joint	1	-	1	
Nuclear	1.99 (1.08–3.66)	0.22	1.69 (0.86–3.30)	0.12
Marital status				
Single	1	-		
Married	1.15 (0.28–4.72)	0.84		
Separated	0.57 (0.09–3.38)	0.53		
Education level				
Diploma	1.00 (0.56–1.77)	0.98		
Bachelors	0.83 (0.43–1.59)	0.57		
Masters	1	-		
Teaching experience (years)				
<1	1.27 (0.61–2.62)	0.51		
1–5	1.15 (0.65–2.03)	0.62		
5–10	0.97 (0.47–1.98)	0.94		
More than 10	1	-		
Status of work				
Permanent	1	-	1	
Temporary	0.27 (0.10–0.74)	0.01	0.42 (0.14–1.26)	0.12
School type				
Government	1	-		
Aided	0.71 (0.39–1.28)	0.25		
Private	0.91 (0.32–2.55)	0.86		
School area				
Rural	1	-	1	
Urban	4.06 (1.67–9.84)	0.002	3.40 (1.22–9.44)	0.01*
Teaching hours per week				
1–5	1	-	1	
6–10	0.79 (0.35–1.75)	0.56	0.50 (0.21–1.20)	0.12
More than 10	2.01 (1.03–3.91)	0.04	1.63 (0.78–3.39)	0.19
Medium of instruction				
Malayalam	1	-		
English	1.10 (0.60–2.01)	0.74		
The average strength of the class				
<30	1	-	1	
30–40	2.69 (1.63–4.45)	<0.001	2.38 (1.37–4.15)	0.002*
More than 40	0.92 (0.34–2.49)	0.88	0.73 (0.25–2.12)	0.57
Syllabus				
State	1	-		
CBSE	1.13 (0.36–3.55)	0.82	-	

*P<0.05 statistically significant. COR: Crude odds ratio, AOR: Adjusted odds ratio, CI: Confidence interval, CBSE: Central Board of Secondary Education

a moderate level of knowledge regarding signs and symptoms of ADHD, but low knowledge related to general features and treatment of ADHD.[28] This study result is consistent with the findings of the present study.

In the current study, very few participants had undergone training programmes related to childhood mental health issues. It was also observed that the participants who had received formal training on childhood mental health issues were able to identify ADHD as compared to the teachers who did not attend training. The current study findings align with a study conducted by Shetty and Rai in 2014. Teachers who had prior training on ADHD were better able to identify ADHD than

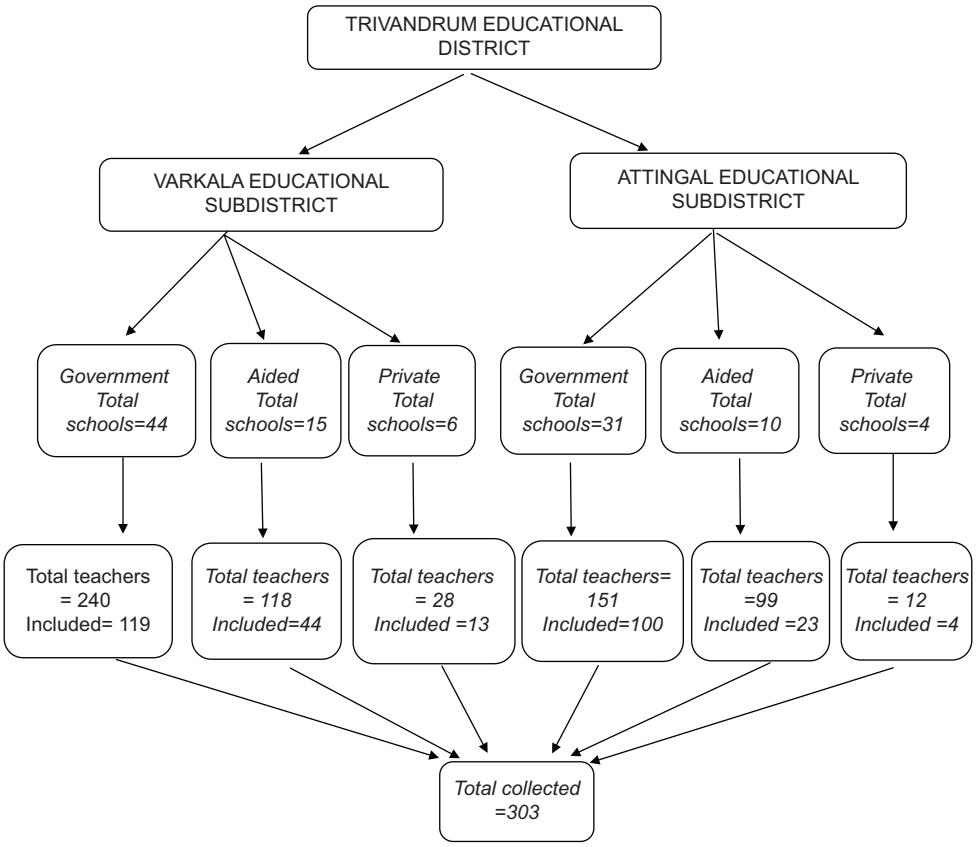


Figure 1: Sampling Design of the study

those who did not undergo training.[29] This suggests that mental health training programmes for primary schoolteachers on childhood mental health issues are the need of the hour. The SDG-3 and SDG-4 also focus on improving teachers' MHL and highlighted the importance of enhancing students' mental health and well-being in educational settings.[30,31]

On comparison of the mean knowledge scores and the socio-demographic characteristics of participants, it was found that marital status, status of work, type of school, teaching hours per week, medium of instruction and average strength of class are statistically significant differences in the mean knowledge score. The study conducted by Sharma and Kunwar in Nepal also found the association between knowledge of ADHD and duration of teaching.[32]

On assessing the predictors of MHL, it was observed that teachers from urban areas had better MHL on ADHD as compared to teachers from rural areas. This might be because overall health literacy in urban populations is higher than that of rural populations.[33] It was also found out that teachers who taught the classes with a strength of 30–40 had better identification of ADHD as compared to teachers who taught smaller class strength. This might be because teachers can more effectively engage with their students without being overwhelmed when the class sizes are moderate.

Strength and limitations of the study

This study is one of the few in India, and specifically in Kerala, to assess ADHD-related MHL amongst primary schoolteachers, thereby addressing a major gap in the literature. A multi-method approach was employed, incorporating validated tools such as the KADDS, case vignettes and behavioural indicators, which enabled a comprehensive assessment of MHL beyond knowledge alone. Furthermore, the relatively large sample size (n = 303) with diverse socio-demographic and school-related characteristics enhanced the representativeness and robustness of the findings.

Limitations

The cross-sectional design of the study limits the ability to infer causal relationships between teacher characteristics and MHL levels. In addition, since the study was conducted in a single state (Kerala), the findings may have limited generalisability to other regions of India, where teacher demographics and cultural contexts may differ.

Considerations for future studies

Future research could use longitudinal or interventional designs to assess the impact of teacher training programmes on ADHD-related MHL. Incorporating qualitative methods, such as interviews or focus groups, may provide deeper insights into teachers' attitudes, cultural perceptions, and challenges in supporting children with ADHD. In addition, studies should explore the effectiveness of policy-level interventions, including the integration of ADHD and MHL training into pre-service and in-service teacher education curriculum.[34]

Conclusion

The MHL regarding ADHD among primary schoolteachers is low. The teachers who did not attend formal training related to mental health issues were not able to identify ADHD. This highlights the importance of incorporating formal training programmes for primary schoolteachers regarding childhood mental health issues. Early identification of children with mental health issues and timely referral are essential to ensure appropriate interventions and support. The findings emphasise the importance of integrating ADHD and broader mental health training into both pre-service and in-service teacher programmes.

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Conflicts of interest

There are no conflicts of interest.

References

1. Núñez-Jaramillo L, Herrera-Solís A, Herrera-Morales WV. ADHD: Reviewing the causes and evaluating solutions. J Pers Med 2021;11:166.
2. Attention-Deficit/Hyperactivity Disorder (ADHD) in Children – Symptoms and Causes. Mayo Clinic; 2019. Available from: https://www.mayoclinic.org/diseases-conditions/adhd/symptoms-causes/syc-20350889. [Last accessed on 2024 May 10].
3. Attention Deficit Hyperactivity Disorder (ADHD). WebMD; 2024. Available from: https://www.webmd.com/add-adhd/attention-deficit-hyperactivity-disorder-adhd. [Last accessed on 2024 May 05].
4. Wilens TE, Spencer TJ. Understanding attention-deficit/hyperactivity disorder from childhood to adulthood. Postgrad Med 2010;122:97-109.
5. Ortiz D. Millions of Indian Children (and Parents) Struggle with ADHD. Available from: https://www.neeuro.com/blog/adhd-india. [Last accessed on 2024 Mar 10].
6. Ayano G, Demelash S, Gizachew Y, Tsegay L, Alati R. The global prevalence of attention deficit hyperactivity disorder in children and adolescents: An umbrella review of meta-analyses. J Affect Disord 2023;339:860-6.
7. Hass M, Ardell A. Supporting Student Mental Health: Essentials for Teachers. 1st ed. Abingdon-on-Thames, UK: Routledge; 2022.
8. Hartman CA, Rhee SH, Willcutt EG, Pennington BF. Modeling rater disagreement for ADHD: Are parents or teachers biased? J Abnorm Child Psychol 2007;35:536-42.
9. United Nations. Sustainable Development Goals (SDG 4). United Nations Western Europe; 2020. Available from: https://unric.org/en/sdg-4/. [Last accessed on 2024 Apr 10].
10. Magna E. Review of Mental Health Policy and Implementation in Ghana: A Roadmap To Achieving Sustainable Development Goal (SDG) 3; 2018. Available from: https://www.academia.edu/37229302/A_REVIEW_OF_MENTAL_HEALTH_POLICY_AND_IMPLEMENTATION_IN_GHANA_A_ROADMAP_TO_ACHIEVING_SUSTAINABLE_DEVELOPMENT_GOAL_SDG_3. [Last accessed on 2024 Apr 15].
11. Al-Moghamsi EY, Aljohani A. Elementary school teachers' knowledge of attention deficit/hyperactivity disorder. J Family Med Prim Care 2018;7:907-15.
12. Frigerio A, Montali L, Marzocchi GM. Italian teachers' knowledge and perception of attention deficit hyperactivity disorder (ADHD). Int J Sch Educ Psychol 2014;2:126-36.
13. Parkavi R, Praveena D, Prithish Bhavan R, Ramya K, Sibin RA. A systematic review and meta-analysis of the prevalence of ADHD among children in India in recent years. Int J Homoeopath Sci 2025;9:582-4.
14. Alshehri AM, Shehata SF, Almosa KM, Awadalla NJ. Schoolteachers' knowledge of attention-deficit/hyperactivity disorder—current status and effectiveness of a knowledge improvement program: A randomized controlled trial. Int J Environ Res Public Health 2020;17:5605.
15. Kurian LR, Shaji A, Alex AM, Rini VB Sr., Chacko SM, George SP. A descriptive study to assess the knowledge on attention deficit hyperactivity disorder (ADHD) among primary school teachers in selected schools, Kottayam. Res J Sci Technol 2022;14:30-6. doi:10.52711/2349-2988.2022.00004.
16. Phoa PKA, Razak AA, Kuay HS, Ghazali AK, Rahman AA, Husain M, *et al*. Predictors of mental health literacy among parents, guardians, and teachers of adolescents in West Malaysia. Int J Environ Res Public Health 2023;20:825.
17. Jorm AF. Mental health literacy. Public knowledge and beliefs about mental disorders. Br J Psychiatry 2000;177:396-401.
18. Parimala L, Timple K. Assess the knowledge on attention deficit hyperactivity disorder among primary school teachers. Int J Eng Res Gen Sci 2019;7:102-6. Available from: https://www.researchgate.net/publication/337331836_ASSESS_THE_KNOWLEDGE_ON_ATTENTION_DEFICIT_HYPERACTIVITY_DISORDER_AMONG_PRIMARY_SCHOOL_TEACHERS. [Last accessed on 2025 Nov 20].
19. Sciutto MJ, Terjesen MD, Bender Frank AS. Teachers' knowledge and misperceptions of attention-deficit/hyperactivity disorder. Psychology in the Schools 2000;37:115-22.
20. Alzahrani MM, Alghamdi AA, Alghamdi SA, Alotaibi RK. Knowledge and attitude of dentists towards obstructive sleep apnea. Int Dent J 2022;72:315-21.
21. The Jamovi Project. Jamovi (Version 2.5) [Computer software]. 2024. Available from: https://www.jamovi.org [Last accessed on 2024 Mar 10].
22. Bursac Z, Gauss CH, Williams DK, Hosmer DW. Purposeful selection of variables in logistic regression. Source Code Biol Med 2008;3:17.
23. Elwardany S, Mohammed M, Ahmed ZA. Teachers' perception and attitudes toward attention deficit hyperactivity disorder in primary schools at Assiut City. ResearchGate; 2015. Available from: https://www.researchgate.net/publication/334611863_TEACHERS_PERCEPTION_AND_ATTITUDES_TOWARD_ATTENTION_DEFICIT_HYPERACTIVITY_DISORDER_IN_PRIMARY_SCHOOLS_AT_ASSIUT_CITY. [Last accessed on 2024 Mar 19].
24. Ma F, Anto MM. Importance of mental health awareness among school teachers in bridging mental health treatment gap in India. Int J Soc Psychiatry 2022;68:233-6.
25. Kashyap RS, Sahithya BR, D'Souza L. Mental health, stigma, resilience, and mental health literacy among schoolteachers: A cross-sectional survey from South India. J Ment Health Hum Behav 2024;29:24-31.
26. Bhattarai L, Sharma M. Knowledge regarding attention deficit hyperactivity disorder of children among schoolteachers at Lalitpur. J Chitwan Med Coll 2019;9. Available from: https://www.nepjol.info/index.php/JCMC/article/view/26905. [Last accessed on 2024 Apr 10].
27. Perold M, Louw C, Kleynhans S. Primary school teachers' knowledge and misperceptions of attention deficit hyperactivity disorder (ADHD). S Afr J Educ 2010;30:457-73.
28. Saad S, Aljanahi F, Coumaravelou S, Agha A, Alsamiri M, Allami S. Knowledge about attention-deficit/hyperactivity disorder among primary schoolteachers in Sharjah, UAE. J Educ Health Promot 2022;11:99.
29. Shetty A, Rai S. Awareness and knowledge of attention deficit hyperactivity disorders among primary school teachers in India. ResearchGate; 2014. Available from: https://www.researchgate.net/publication/279187929_Awareness_and_Knowledge_of_Attention_Deficit_Hyperactivity_Disorders_Among_Primary_School_Teachers_in_India. [Last accessed on 2024 Mar 27].
30. United Nations. The Sustainable Development Goals Report 2023.

New York: United Nations; 2023. Available from: https://unstats.un.org/sdgs/report/2023/. [Last accessed on 2025 Sep 09].
31. United Nations Educational, Scientific and Cultural Organization. Teachers and Education for Sustainable Development Goals. Paris: United Nations Educational, Scientific and Cultural Organization; 2022. Available from: https://unesdoc.unesco.org/ark:/48223/pf0000380219. [Last accessed on 2025 Sep 09].
32. Sharma P, Kunwar P. Awareness on attention deficit hyperactivity disorder among primary school teachers of public schools in Kathmandu. J Health Allied Sci 2022;11:28-33. Available from: https://www.jhas.org.np/jhas/index.php/jhas/article/view/214. [Last accessed on 2024 Mar 10].
33. Aljassim N, Ostini R. Health literacy in rural and urban populations: A systematic review. Patient Educ Couns 2020;103:2142-54.
34. Prabhu V, Ashok L, Kamath VG, Puranik A, Padickaparambil S, Rebello NS, *et al*. Evaluating the effectiveness of a mental health literacy module: A randomized controlled trial among school teachers. Clin Epidemiol Glob Health 2025;32:101946.